Sr.No. 4607

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:202 Course Section:1	Seat No: 2022600013

1_61_471_986_1_1505_48284

Name of the candidate. Param Gogia

Candidate's UID number: 2 0 2 2 6 0 0 0 1 3

Examination class room number: 2 0 2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE AIML

DAY AND DATE: 8/10/2025 Wednesday

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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1a ① $f(w) = \frac{1}{2} \|(Xw - y)\|^2$

partially derivating the above equation wrty

$$\frac{\partial f(w)}{\partial y} = -\frac{2}{2}(xw - y)$$

$$\frac{\partial f(w)}{\partial y} = -(xw - y)$$

② Basedon this the gradiant updation formula for this particular loss function canbe obtained.

③ Now, during backpropagation, the gradiant descent can be added to the weights with appropriate learning rate n, thereby weight updation formula will be

$$w_n = w_0 - n \frac{\partial f(w)}{\partial y}$$

$$w_n = w_0 - n\,(xw - y)$$

④ Now for the following gradiants to converge let w^* be the optimal weights required.

⑤ We know fora real way updation of gradiant descent let k be the constant and thereby for all learning weights.

$$(w^T w^*) \leq k\gamma$$

$$\text{\&}\ (w^T w^*) \geq k\gamma$$

~~for~~ By using Cauchy swartz

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inequality

$$|k\gamma^4 \leq |k\gamma|$$

which implie

$$k \leq \frac{R^2}{\gamma}$$

for all possible real values,

Let λ_{max} be the largest possible value of ~~$w^* w$~~ $w^T w^*$ ~~for eig these vectors.~~

for eight valule calculation $\frac{\partial f}{\partial y}$ tend to zero and baud on above calculation, for η being the learning rate in gradient updation formula.

$$\gamma K \leq \frac{R^2}{\gamma}$$

~~derivating on~~

~~dub~~ derivation on both the sides & eliminating constant we get

$$\eta < \frac{2R}{\gamma}$$

$$\eta < \frac{2}{\lambda_{max}}$$

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1b ① Bias-variance decomposition for any model can be shown as follows.

value = $bias^2$ + variance + h

② Now, during the decomposition for expected square predictor, high bias indicates underfitting while low bias & high variance would indicate overfitting.

③ ~~generalization of the decomposition function of a high capacity model leads to overfi~~

③ During overfitting, the variance factor increments exponentially in a high capacity model.

④ Even at generalization of the framework, the overfitting would not eliminate as the variance remains the same, while ~~bis~~ bias can be reduced.

⑤ Here, high capacity models can be regularized using L1 & L2 regularisation thereby affecting the accuracy and not causing overfitting

⑥ In this framework, every time we update the value of square error post regulation, the overfitting model capacity shows suboptimal behaviour in variance calculation.

⑦ Thereby normal behaviour can be obtained.

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2a

① for a given standard convolution of input size H x W x C the number of computations can be calculated as follows.

$$N = H \times W \times \cancel{C} \times K \times F$$

② Now, for a given depthwise seperable autoencoder, the number of convolutions increment thereby the number of computations increase; So, n be the number of convolutions.

$$n = H \times W \times C \times K \times f$$

③ In depthwise seperable convolutions, the flattened vector is important prior that contains the sparse features shared by convolutions at various depth where the dense network of neurons trained can predict output accurately.

④ In modern algorithms for image classification, object segmentation in realtime, the CNN must be ~~d~~ trained to segregate the classes of object and classify them efficiently.

⑤ Prime examples can be tesla auto driving car, where depth wise CNN segregates the objects appearing in front of the car efficiently.

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6) Number of computations in depthwise ~~auto~~ seperable convolutions, help to enhance the output however it surges the parameters to be passed past convolution in the neural network and may decrease the overall efficiency of convolution.

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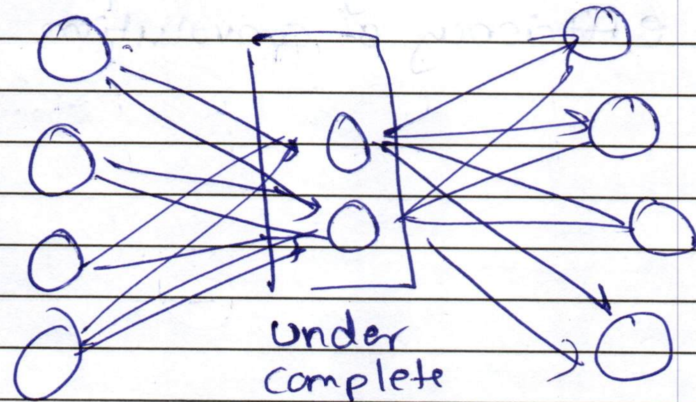
2b ① A linear under complete autoencoder contains lesser number of neurons in the bottleneck layer than the input layer with a single layer between.



② Here, in a linear undercomplete layer, the latent space generated on training with mean square error is of lesser dimension than the overcomplete /sparse/ depth/~~denoised~~ auto encoders.

③ Now, for the autoencoder ~~of~~ trained with MSE,

$$L = \frac{1}{N} \sum_{i=0}^{n} (y-\tilde{y})^2$$

$$\frac{\partial L}{\partial \tilde{y}} = (y-\tilde{y})$$

④ This corresponds to the Loss function of the eigenvalue composition of a subspace with greater dimension.

⑤ Here the lower dimensional lantent space is generated due to lesser parameters that are trained with MSE thereby can equivalent to ~~bd~~ PCA.

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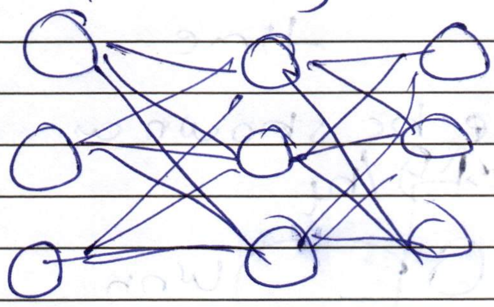
(6) An overcomplete auto encoder contains the number of bottleneck neurons equivalent to the input layer.

(7) without regularisation the all the neuron fire simultaneously and learns the input as it is thereby the output generated would be same as that of the input.

(8) This is highly dangerous as the purpose of encoder, decoder architecture vanishes, thereby becoming an identity fn



(9) In order to avoid this, the bottleneck neurons are penalised inorder to effectively train the neural network shutting of some of them, this process is called regularisation.

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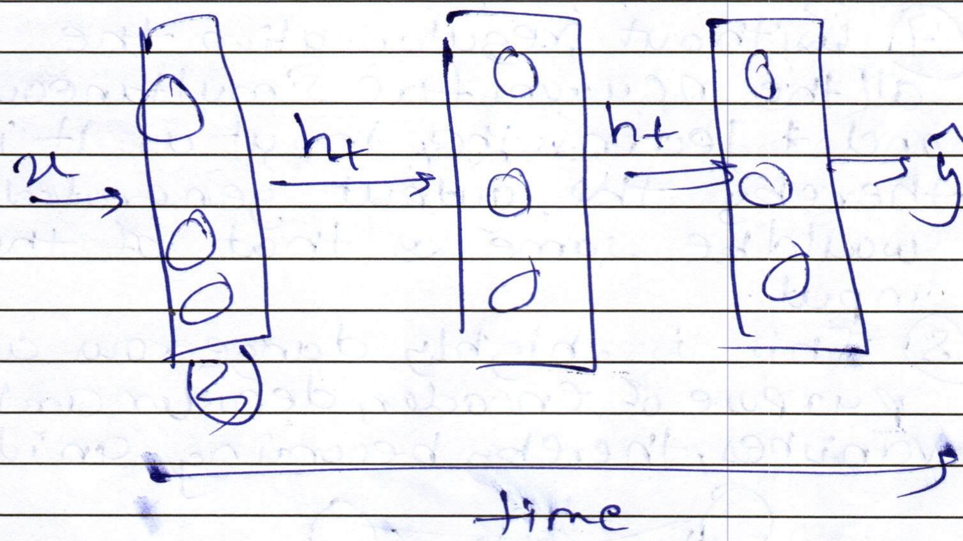
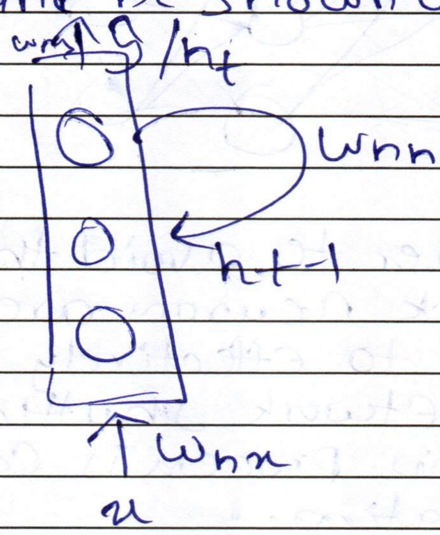
3a

(1) In an ~~nn~~ RNN, the weight updation, output and loss calculation can be shown below



which cane be shown as



$$h_t = W_{hx}x + W_{hh}h_{t-1} + b$$

$$y = W_{hy}h_t + b$$

$$L = \frac{1}{2}(y-\hat{y})^2$$

(2) Over time the loss function gradiant increases the vanishing gradiant problem due to its minimalistic values for past inputs.

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(3) partially differentiating the Loss function wrt to h_t of k period

$$\frac{\partial L}{\partial h_{t-k}} = 2\frac{\partial (y-\hat{y})}{\partial h_{t-1}}$$

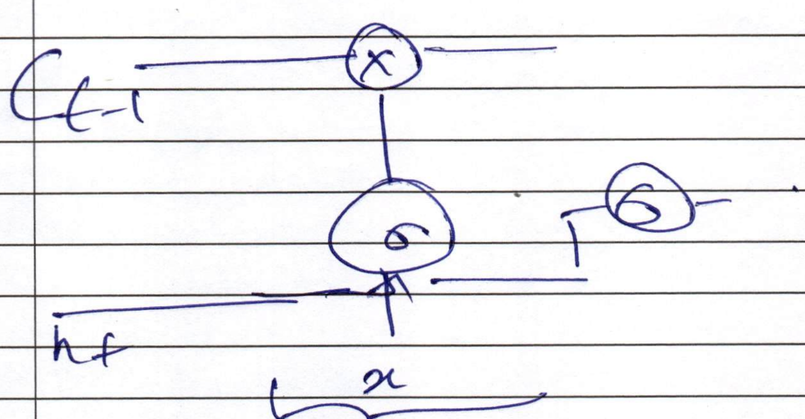
$$= \frac{\partial (w_{hy} x h_t + b)}{\partial h_{t-k}}$$

- let k be the constant

$$= k\frac{\partial h_t}{\partial h_{t-1}} \times \frac{\partial h_{t-1}}{\partial h_{t-2}} \times \frac{\partial h_{t-2}}{\partial h_{t-3}} \times \ldots$$

$$= \frac{\partial h_t}{\partial h_{t-1}} \times \frac{\partial h_{t-1}}{\partial h_{t-2}} \cdots \times \frac{1}{\partial h_{t-k}}$$

(4) In LSTM, the forget equation can be shown below with architecture



forget gate

$$f = \sigma(W_{hf} x + b_f)$$

when updating the cell state

$$\hat{C} = f(C_{t-1}) + i(\hat{C})$$

the forget gate values are taken into considerati

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(5) Forget gate enables the LSTM to not to take insto consideration the current input value or forget the past value for the current parameter
(6) This increases the contextual awareness of the model while prediction thereby eliminating the varishing gradient problem

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3b) 1) The gating equations of GRU can be shown below

(i) Update gate

$$z = \sigma(W_{zh} x + b_z)$$

(ii) Reset gate

$$r = \sigma(W_{rh} x + b_r)$$

(ii) Hidden state cell ~~updation~~ (candidate)

$$\hat{z} = \tanh(r * W_{zh} x + b)$$

(iii) State cell updation.

$$h_t = \hat{z}\, z + (1-\hat{z}) r$$

2) Here this is a combination of update gate with candidate along with reset gate with the $(1-\hat{z})$ of the candidate

3) Thereby while updation the previous cell state with the current is ~~ka~~ taken into consideration.

4) Mathematically, LSTM comprises of 3 gates. → forget gate, input gate and output gate where 3 parameters / o/p of these gates are taken into consideration while updating the hidden state with current $(f, i, o, \underset{\uparrow}{c})$.

candidate

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(5) ~~In~~ In GRU the parameters are Update gate, reset gate, candidate then ~~then~~ three mathematical parameters contribute to the ~~State~~ hidden state updation thereby eliminating the redundant gate required, leading to faster o/p calculation.

(6) Generalisation can improve the use of these models in real world sentence translation, next word prediction and longer context awareness.

(7) This would enhance the accuracy of the models and make it a ~~very~~ efficient to large number of NLP applications.

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